

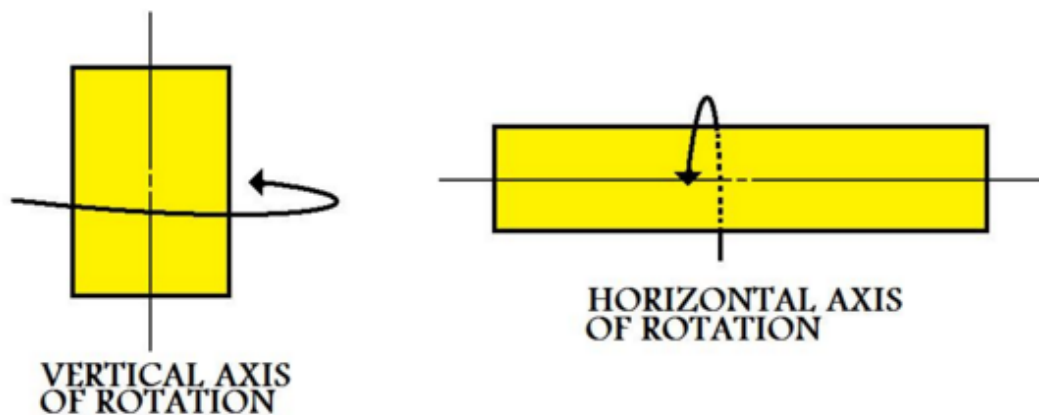
True Centrifugal Casting

The Process

It is necessary when manufacturing a cast part by the true centrifugal metal casting process, using some mechanical means, to rotate the mold. When this process is used for industrial manufacture, this is accomplished by the use of rollers. The mold is rotated about its axis at a predetermined speed. Molds for smaller parts may be rotated about a vertical axis. However, most times in true centrifugal casting manufacture the mold will be rotated about a horizontal axis. The effects of gravity on the material during the metal casting process make it particularly necessary to cast longer parts with forces generated from horizontal rather than vertical rotation.

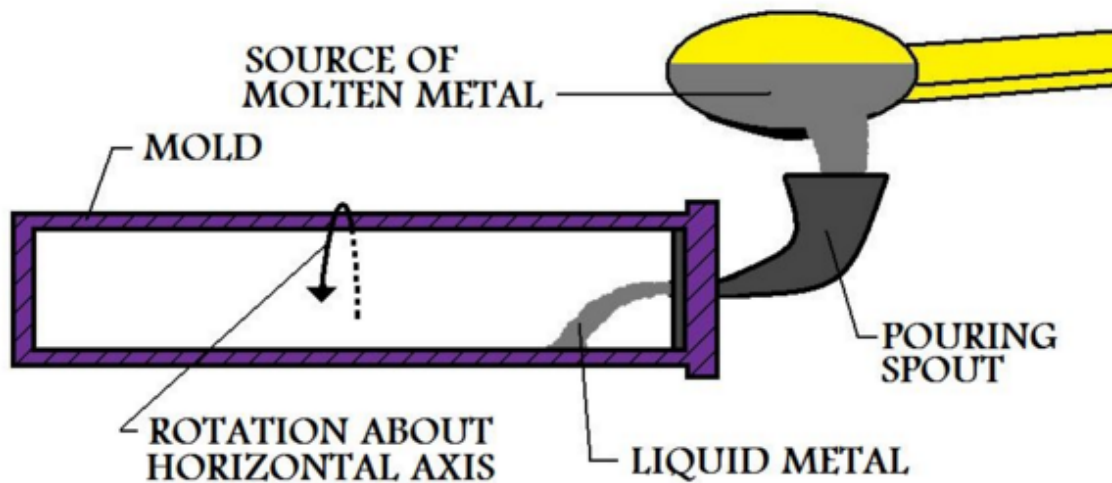
Figure:91

VERTICAL VS HORIZONTAL ROTATION

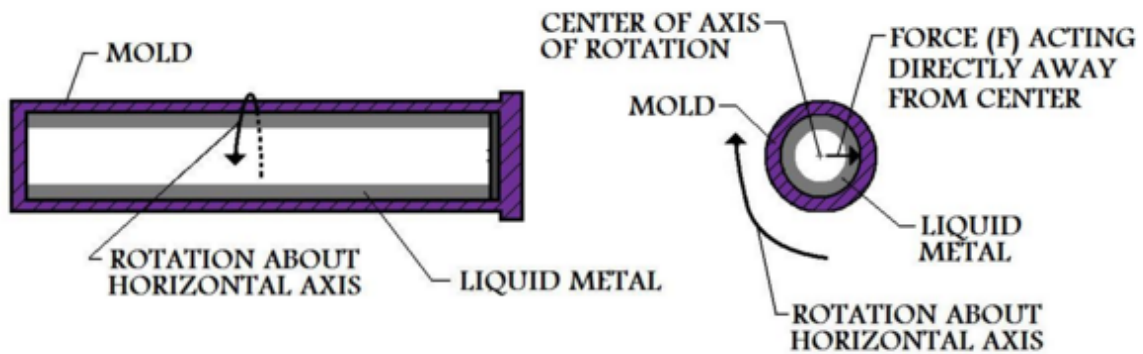


The molten material for the cast part is introduced to the mold from an external source, usually by means of some spout. The liquid metal flows down into the mold. Once inside the cavity, the centripetal forces from the spinning mold force the molten material to the outer wall. Molten material for the casting may be poured into a spinning mold or the rotation of the mold may begin after pouring has occurred.

POURING IN TRUE CENTRIFUGAL CASTING



SOLIDIFICATION IN TRUE CENTRIFUGAL CASTING



It can be seen that this casting process is very well suited for the manufacture of hollow cylindrical tubes. The forces used in this technique guarantee good adhesion of the casting material to the surface of the mold. Thickness of the cast part can be determined by the amount of material poured. The outer surface does not need to be round. Polygonal geometries such as squares and other shapes can be cast. However, due to the nature of the process, the inner surface of a part manufactured by true centrifugal casting must always be round.

Properties And Considerations Of Manufacturing By True Centrifugal Casting

- True centrifugal casting is a great manufacturing process for producing hollow cylindrical parts.
- The metal casting's wall thickness is controlled by the amount of material added during the pouring phase.
- Rotational rate of the mold during the manufacture of the casting must be calculated carefully based on the mold dimensions and the metal being cast.
- If the rotational rate of the mold is too slow, the molten material for the casting will not stay adhered to the surface of the cavity. From the top half of the rotation it will rain metal within the casting cavity as the mold spins.
- This manufacturing operation produces metal cast parts without the need for sprues, risers, or other gating system elements, making this a very efficient industrial metal casting process, in terms of material usage.
- Since large forces press the molten material for the cast part against the mold wall during the manufacturing operation, good surface finish and detail are characteristic of true centrifugal casting.
- Quality castings with good dimensional accuracy can be produced with this process.
- Material of high density and with few impurities is produced in the outer regions of cylindrical parts manufactured by true centrifugal casting.

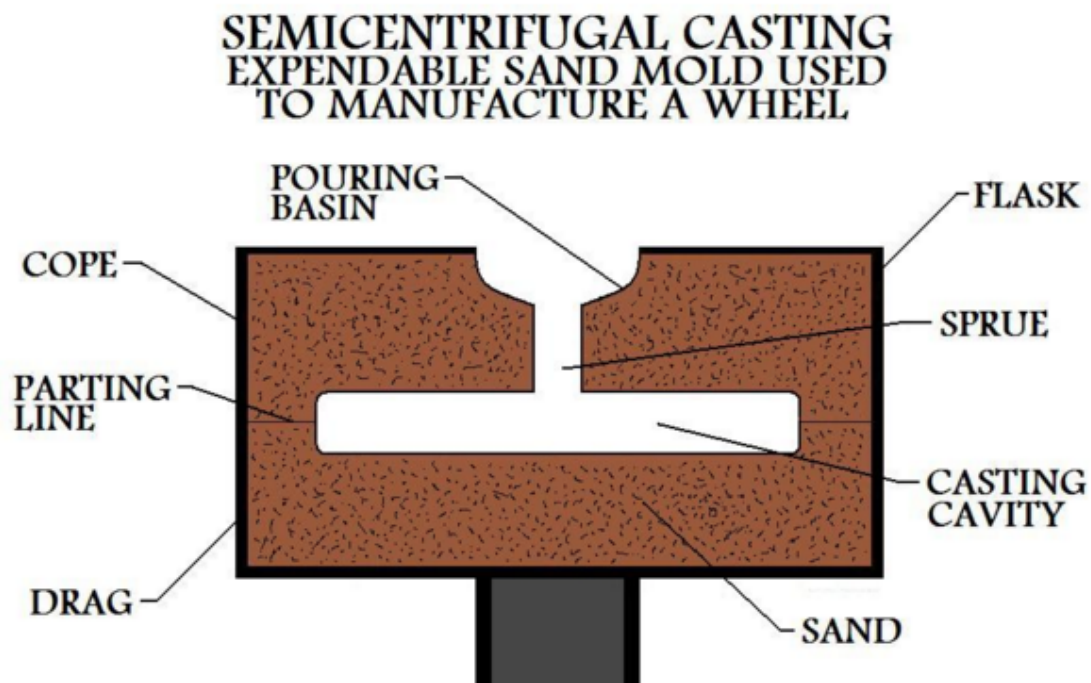
- Impurities, such as metal inclusions and trapped air, collect in the lower density inner regions of cylindrical parts cast by this process.
- These inner regions can be machined out of the cast part leaving only the dense, more pure material.
- Shrinkage is not a problem when manufacturing by true centrifugal casting, since material from the inner sections will constantly be forced to instantly fill any vacancies that may occur in outer sections during solidification.
- This method can produce very large metal castings. Cylindrical pipes 10 feet in diameter and 50 feet long have been manufactured using this technique.
- With the employment of a sand lining in the mold, it is possible to manufacture castings from high melting point materials such as iron and steels.
- This is a large batch production operation.
- True centrifugal casting is a manufacturing process that is capable of very high rates of productivity.

Semicentrifugal Casting

Semicentrifugal casting manufacture is a variation of true centrifugal casting. The main difference is that in semicentrifugal casting the mold is filled completely with molten metal, which is supplied to the casting through a central sprue. Castings manufactured by this process will possess rotational symmetry. Much of the details of the manufacturing process of semicentrifugal casting are the same as those of true centrifugal casting. For a better understanding of this process and centrifugal casting manufacture in general see [true centrifugal casting](#). Parts manufactured in industry using this metal casting process include such things as pulleys, and wheels for tracked vehicles.

In semi-centrifugal casting manufacture a permanent mold may be employed. However, often industrial manufacturing processes will utilize an expendable sand mold. This enables the casting of parts from high temperature materials.

Figure:96

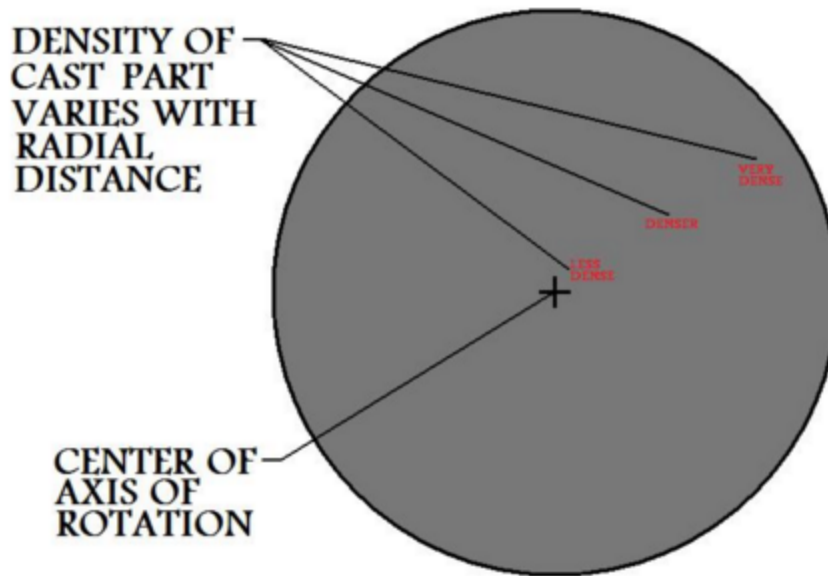


The molten material for the metal casting is poured into a pouring basin and is distributed through a central sprue to the areas of the mold. The forces generated by the rotation of the mold ensure the distribution of molten material to all regions of the casting.

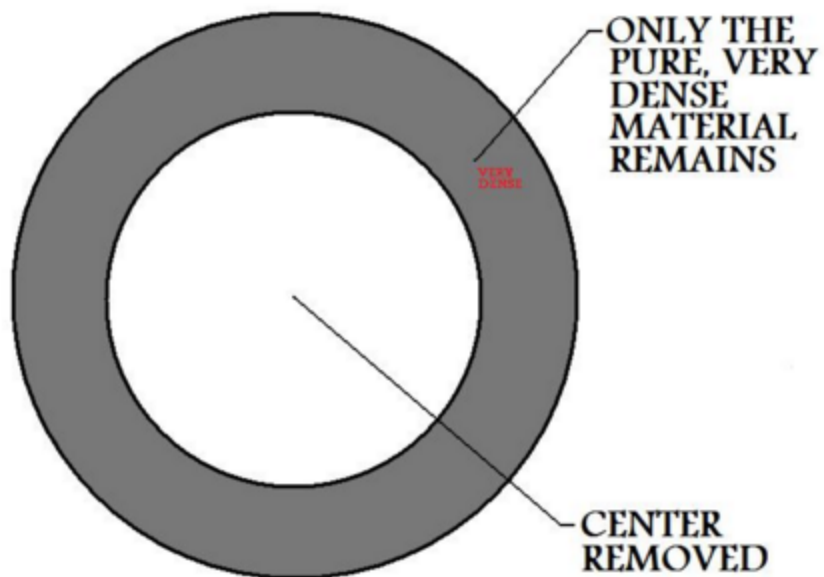
The centripetal forces acting on the casting's material during the manufacturing process of semicentrifugal casting, play a large part in determining the properties of the final cast part. This is also very much the case with cast parts manufactured using the true centrifugal casting process. Forces acting in the true centrifugal process are similar to those that influence the material of a metal casting being manufactured by semicentrifugal casting.

When manufacturing by semi-centrifugal casting, the centripetal acceleration generated on the mass of molten metal by a rotating mold is the force that acts to fill the casting with this molten metal. This is also the force that continues to act on the material as the casting solidifies. The main thing to remember about centripetal forces is that the force will push in a direction that is directly away from the center of the axis of rotation.

CAST WHEEL



CAST WHEEL



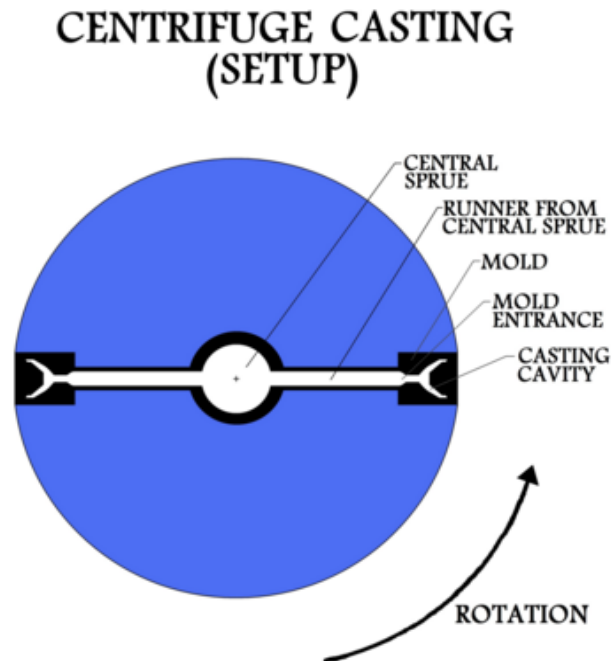
CENTRIFUGE CASTING

processes. Centrifuge casting is different in that castings manufactured by the centrifuge casting process need not have rotational symmetry. With centrifuge casting, metal castings of desired shapes can be manufactured with all the distinct benefits of castings produced by a centrifugal casting process.

The Process

In centrifuge casting manufacture, molds employed to produce the desired castings are arranged around a central sprue. These molds contain all the necessary geometry for the cast part, as well as the gating system. Runners travel from the central sprue to the mold entrances. Runners travel from the central sprue to the mold entrances.

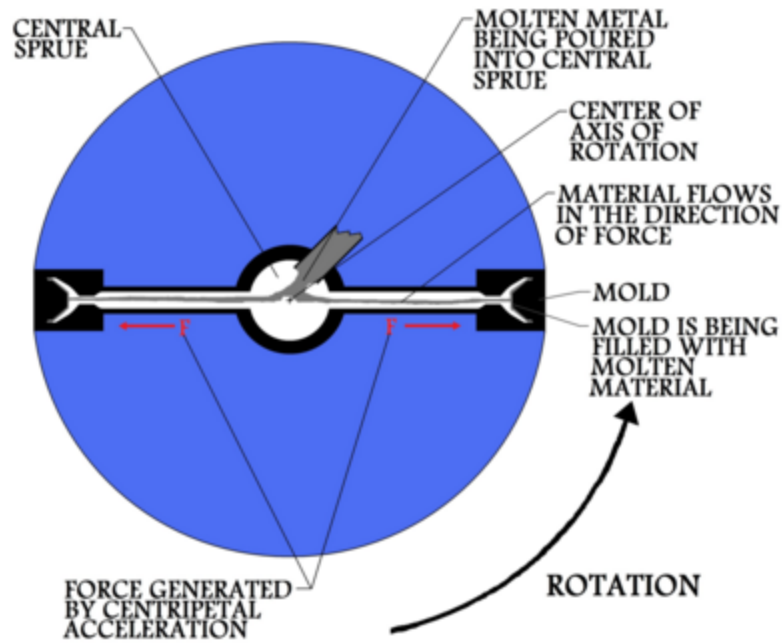
Figure:105



During the pouring phase of centrifuge casting manufacture, molten material is introduced into the central sprue. The entire system is rotated about an axis with the central sprue at the center of rotation. When an object is rotated, forces are produced that act directly away from the center of the axis of rotation. It would be known from the previous discussions concerning the other two branches of centrifugal casting, that the utilization of the forces of centripetal acceleration which act to push material away from the center of rotation is the trademark characteristic of all the manufacturing processes of centrifugal casting. Centripetal force is not only utilized to distribute molten material through a mold, but to help control the material properties of a cast part.

In centrifuge casting manufacture, the molten material to produce the casting is poured into the central sprue. Centripetal forces from the rotating apparatus push this material outward from the center, through the runners and into the molds.

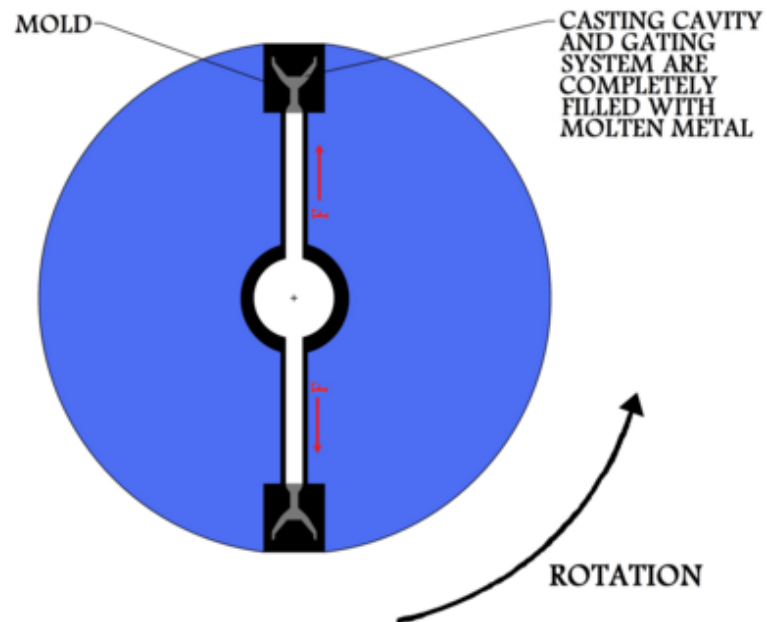
CENTRIFUGE CASTING (POURING)



When the correct amount of molten metal to manufacture the casting is poured and distributed completely into the molds, the apparatus will continue to rotate as solidification is occurring.

Figure:107

CENTRIFUGE CASTING (SOLIDIFICATION)



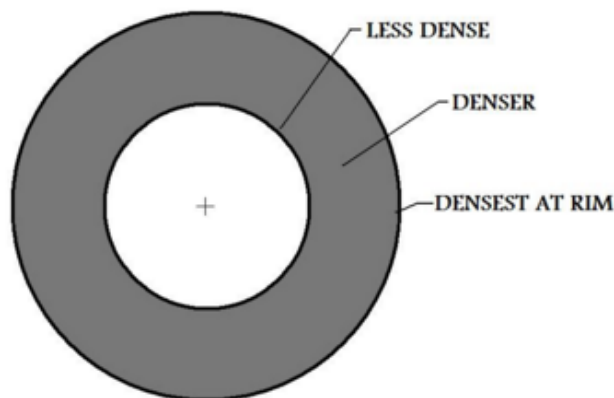
Centrifuge Casting Manufacture

There are many specific advantages in the quality of parts manufactured by centrifuge casting. Since the metal is forced into the mold, the mold cavity usually fills completely and cast parts with thin walled sections are possible. Also, great surfaces can be produced by centrifugal casting, which is another characteristic of castings manufactured by a process that uses large amounts of force to fill a mold.

One of the most notable features, specific to the centrifugal casting processes and discussed in the preceding sections, is the effect of centripetal forces acting continuously on the material as the casting solidifies. Molten material that solidified under greater force will be denser than the same material that solidified under less force. This can be observed in a round cylinder manufactured using the [true centrifugal casting](#) process.

Figure:108

CAST PART PRODUCED BY TRUE CENTRIFUGAL CASTING



Continuous Casting

Continuous casting, also referred to as strand casting, is a process used in manufacturing industry to cast a continuous length of metal. Molten metal is cast through a mold, the casting takes the two dimensional profile of the mold but its length is indeterminate. The casting will keep traveling downward, its length increasing with time. New molten metal is constantly supplied to the mold, at exactly the correct rate, to keep up with the solidifying casting. Industrial manufacture of continuous castings is a very precisely calculated operation. Continuous casting can produce long strands from aluminum and copper, also the process has been developed for the production of steel.

The Process

Molten metal, from some nearby source, is poured into a **tundish**. A tundish is a container that is located above the mold, it holds the liquid metal for the casting. This particular casting operation uses the force of gravity to fill the mold and to help move along the continuous metal casting. The tundish is where the operation begins and is thus located high above ground level, as much as eighty or ninety feet. As can be seen, the continuous casting operation may require a lot of space.

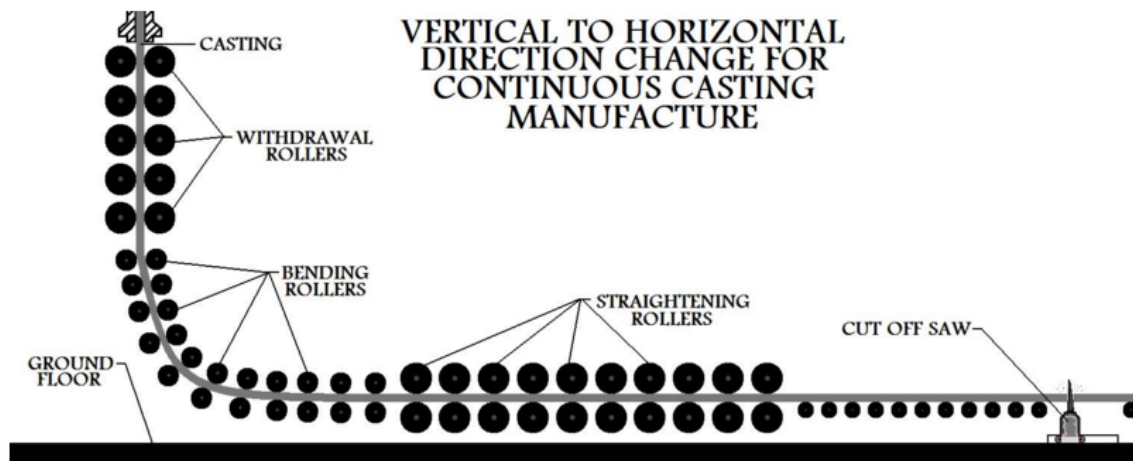
It is the job of the tundish to keep the mold filled to the right level throughout the manufacturing operation. Since the metal casting is constantly moving through the mold, the tundish must always be supplying the mold with more molten metal to compensate.

The supplying of metal to the mold is not only going on throughout the entire manufacturing operation, it must be carried out with accuracy. A control system is employed to assist with this task. Basically the system can sense what the level of molten metal is, knows what the level should be, and can control the pouring of the metal from the tundish to ensure the smooth flow of the casting process. Although the tundish can typically hold several thousand pounds of metal, it too must be constantly supplied from the source of molten material.

The tundish also serves as the place where slag and impurities are removed from the melt. The high melting point and reactive nature, at high temperatures, has always made steel a difficult material to cast. When a manufacturing operation is continuously casting steel, the reactivity of the molten steel to the environment needs to be controlled. For this purpose, the mold entrance may be filled with an inert gas such as argon. The inert gas will push away any other gases, such as oxygen, that may react with the metal. There is no need to worry about the inert gas reacting with a molten metal melt, since inert gases do not react with anything at all.

The metal casting moves quickly through the mold, in the continuous manufacture of the metal part. The casting does not have time to solidify completely in the mold. As can be remembered from our discussion on solidification, a metal casting will first solidify from the mold wall, or outside of the casting, then solidification will progress inward. The mold in the continuous casting process is water cooled, this helps speed up the solidification of the metal casting. As stated earlier, the continuous casting does not completely harden in the mold. It does, however, spend enough time in the water cooled mold to develop a protective solidified skin of an adequate thickness on the outside.

The long metal strand is moved along at a constant rate, by way of rollers. The rollers help guide the strand and assist in the smooth flow of the metal casting out of the mold and along its given path. A group of special rollers may be used to bend the strand to a 90 degree angle. Then another set will be used to straighten it, once it is at that angle. Commonly used in manufacturing industry, this process will change the direction of flow of the metal strand from vertical to horizontal.



The continuous casting can now travel horizontally as far as necessary. The cutting device, in manufacturing industry, is typically a torch or a saw. Since the metal casting does not stop moving, the cutting device must move with the metal casting, at the same speed, as it does its cutting. There is another commonly used setup for cutting lengths of metal casting strand from a continuous casting operation. This particular manufacturing setup eliminates the need for bending and straightening rollers. It does, however, limit the length of metal casting strand that may be produced, based in a large part on the height of the casting floor where the mold is located.

Properties And Considerations Of Manufacturing By Continuous Casting

- Continuous casting manufacture is different from other metal casting processes, particularly in the timing of the process. In other casting operations, the different steps to the process such as the ladling of metal, pouring, solidification, and casting removal all take place one at a time in a sequential order. In continuous casting manufacture, these steps are all occurring constantly and at the same time.
- This process is used in commercial manufacture as a replacement to the traditional process of casting ingots.
- Piping, a common problem in ingot manufacture, is eliminated with the continuous casting process.
- Structural and chemical variations in the metal of the casting, often present in ingots, have been eliminated. When manufacturing with the continuous metal casting process, the casting's material will possess uniform properties.
- When employing continuous metal casting manufacture, the castings will solidify at 10 times the rate that a casting solidifies during ingot production.
- With less loss of material, cost reduction, higher productivity rate, and superior quality of castings, continuous casting manufacture is often the choice over ingot production.
- A continuous casting manufacturing process will take considerable resources and planning to initiate, it will be employed in only very serious industrial operations.